

With standard Ethernet, you can re-flash the car in minutes compared to hours

There are multitudes of networking technologies available, but very few are future-proof for the technological advances that lie ahead. How do you select the perfect solution?

Michael Jones, product marketing director, Micrel, speaks to Dilin Anand and Sneha Ambastha about standard Ethernet for automotive systems, and why it could soon rule the pack



MICHAEL JONES
PRODUCT MARKETING DIRECTOR, MICREL

Could you give us a primer as to how important Ethernet is to electronic systems in the automotive industry?

Ethernet has already been widely accepted by the automotive industry as the preferred interface for on-board diagnostics (OBD) and has been deployed in various car models since 2008. Ethernet provides increased bandwidth speeds over traditional automotive buses, resulting in a reduction in software download times from hours to minutes compared to traditional methods. This adoption of Ethernet has already been standardised in ISO 13400 diagnostics over IP, using Ethernet as the physical layer.

Following success in diagnostics, the industry is looking at Ethernet to provide next-generation solutions for advanced driver assistance systems (ADAS) and in-vehicle infotainment (IVI). ADAS constitutes one of the fastest growing applications within the automotive market, driven by government legislation and a desire for enhanced in-vehicle safety.

Could you give us an example of which specification suits which application the best?

The needs of vehicle infotainment systems (and ADAS) differ from the current applications, such as OBD in that these are applications operating in real time whilst the car is moving. The original Ethernet IEEE802.3 specification was not designed for real-time applications; however, recent Ethernet audio video bridging (AVB) standard addresses the need for real-time A/V applications.

IEEE AVB specifications provide the necessary guaranteed quality of service (QoS) for such AV streaming applications in the car, the home and for professional AV equipment. The standard comprises three key specifications:

1. IEEE 802.1as time synchronisation
2. IEEE 802.1Qat stream reservation
3. IEEE 802.1Qav queuing and forwarding for AV bridges

Could you elaborate on these three specifications?

Time synchronisation is critical in order to ensure quality audio and video streaming within an Ethernet network.

IEEE 802.1as utilises specific precise time protocol (PTP) packets to provide synchronisation across the network to a common system clock source. Nodes in the network, known as time-aware bridges, will extract timing from the network based on a series of PTP synchronisation message exchanges with the master clock source and neighbours. As a consequence, audio and video sources can be synchronously streamed across the vehicle network.

IEEE 802.1Qat stream reservation allows network bandwidth and buffer resources to be reserved for specific traffic schemes using stream reservation protocol (SRP). This ensures a guaranteed QoS for A/V and high-priority traffic, preventing any packet loss or significant network delay during periods of congestion. IEEE 802.1Qav queuing and forwarding methods are based on segregating traffic into isochronous (time critical) and asynchronous (non-time critical) packets and prioritising using the priority class defined in IEEE 802.1p. A credit-based traffic shaper is defined to smooth the 'bursty' nature of video data.

The bandwidth in case of Ethernet would increase to 100Mbps per node or even 1Gbps. What has led to this increase? Is it due to different topologies?

It is partly due to the topology of next-generation networks and also the increased bandwidth of applications such as video – high definition for infotainment and camera imaging for driver assistance, for example, rear-view, lane departure, signpost, traffic light and pedestrian recognition.

Next-generation vehicle networks will take advantage of the additional bandwidth, for example, GigE offered by Ethernet to provide a high-speed backbone in the car connecting various domains.

What is the 'best master clock' algorithm and how does this affect the AVB system?

For IEEE 802.1as time precision protocol (PTP) each network node (slave) is synchronised to a master clock in the network. The best master clock mechanism is a way to select